

CSE 215: Database

Lecture 01

2022

Introductory Lecture

Teaching Staff

- Instructors:



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Lectures: 1st half - from a user's perspective

1. **Foundations:** Relational data models & SQL
 - Weeks 1-3
 - How to manipulate data with SQL, a declarative language
 - *reduced expressive power but the system can do more for you*
2. **Database Design:** Design theory and constraints
 - Weeks 4-5
 - Designing relational schema to keep your data from getting corrupted
3. **E-R diagram & Relational Algebra:** Syntax & supporting systems
 - Week 6-7
 - A programmer's first step to data abstraction

Lectures: 2nd half - understanding how it works

4. Inside to database systems

- Indexing
- External Memory Algorithms (IO model) for sorting, joins, etc.
- Basics of query optimization (Cost Estimates)

5. Transactions: Syntax & supporting systems

- ✓ A programmer's abstraction for data consistency

6. Data Recovery

Communications

- Web site:

<http://teacher.buet.ac.bd/ashikur/CSE215>

- Lectures available here (usually the morning before class)
- Useful links to **possibly** useful reading materials
- Class test marks and important announcements are posted !

- Email:

{ashikur, asmlatifulhoque}@cse DOT buet DOT ac DOT bd

Textbook(s)

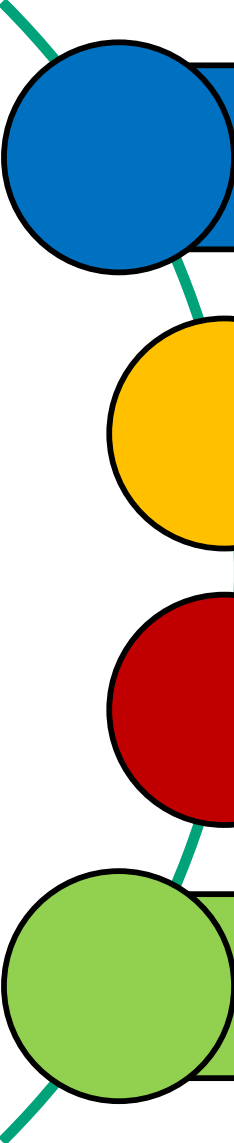
Main textbook:

- *Database Systems: The Complete Book*, Hector Garcia-Molina, Jeffrey Ullman, Jennifer Widom
 - Most chapters are good. Some are not as great (functional dependencies).
 - available at the Poroma Prokashoni
 - A free PDF version is available (linked from the course website)
- COME TO THE CLASS ! ASK QUESTIONS !
READ SLIDES !

Other Texts

- *Database System Concepts*,
by Avi Silberschatz, Henry F. Korth, S. Sudarshan

Today's Outline



Overview of DBMS

DBMS vs other alternatives

DBMS through an example

The Course outline

“Data is the Future”

- Quotation by somebody on the earth

What is a database ?



FACEBOOK
definition

“Data is the Future”

- Quotation by somebody on the earth

- What is a database?
- Any collection of related information
 - Phone book
 - Shopping list
 - Todo list
 - Your 5 best friends
 - Facebook’s user base
- Database can be stored in different ways
 - On paper
 - In your mind
 - On a computer

“Data is the Future”

- Quotation by somebody on the earth

Give examples of databases

- Accounts database; payroll database; BIIS students database



Increasingly many companies
see themselves as **data driven**.

Computers + Databases = <3

Amazon.com

- Keeps track of Products, Reviews, Purchase Orders, Credit Cards, Users, Media, etc
- Trillions of pieces of information need to be stored and readily available
- Information is extremely valuable and critical to Amazon.com's functioning
- Security is essential, Amazon stores peoples personal information
 - Credit card #, SSN, Address, phone
- Information is stored on a computer

vs

Shopping List

- Keeps track of consumer products that need to be purchased
- 10-20 pieces of information need to be stored and readily available
- Information is for convenience sake only and not necessary for shopping
- Security is not important
- Information is stored on a piece of paper, or even just in someone's memory

Millennia of Knowledge in Libraries



The world's scientific knowledge
is **accessible**.

But we're still human...



The world's scientific knowledge
is **accessible**, but not **readable**.

Could we build a machine to
read for us?

Building database is the first step
towards machine readable data

Database Management System

What is a DBMS ?

- *A big C/C++ program written by someone else that allows us to manage efficiently a large database and allows it to persist over long periods of time*

Database Management System

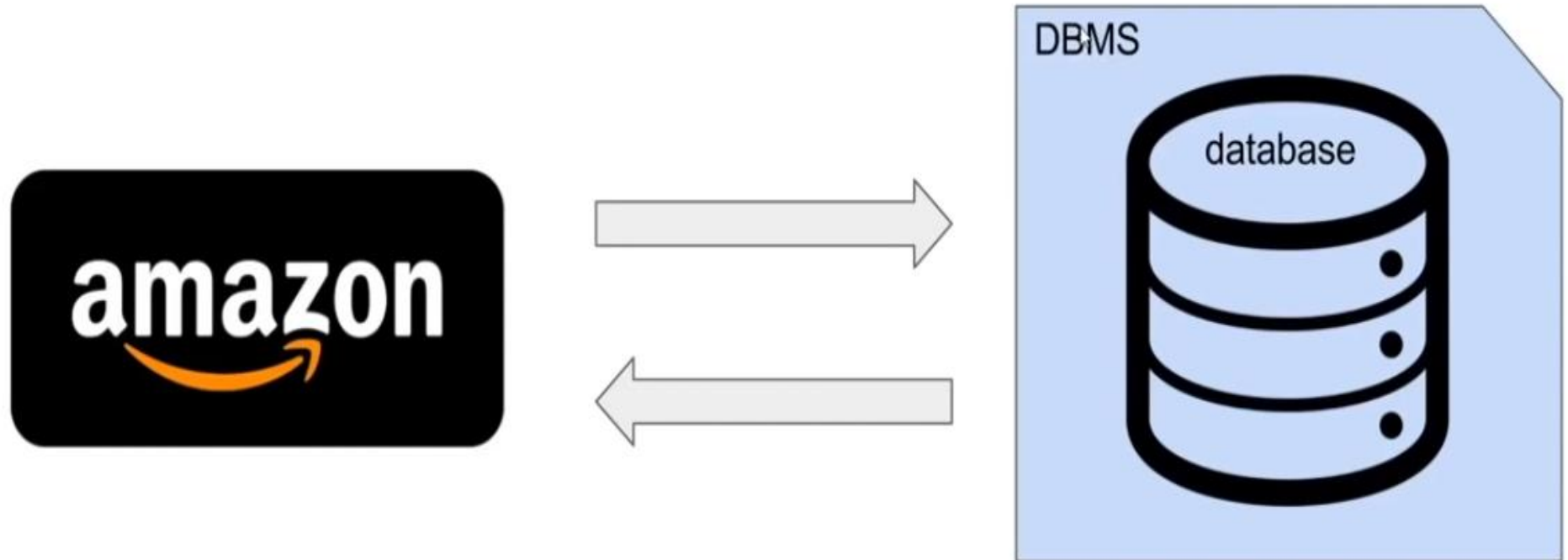
What is a DBMS(formal definition)?

•A **Database Management System (DBMS)** is a piece of software designed to store and manage databases

Functionalities of DBMS

- Makes it easy to manage large amounts of information
- Handles Security
- Backups
- Importing and exporting data
- Concurrency
- Interacts with software application

Amazon.com Database Diagram



Amazon.com will interact with the DBMS in order to create, read, update and delete information

Database Management System

Give examples of DBMS

- The big commercial database vendors:
 - Oracle
 - IBM (with DB2) bought Informix recently
 - Microsoft (SQL Server)
- Some open source (free) database systems:
 - Postgresql
 - Mysql
 - Predator

DBMS Market Shares

In 2006, www.gartner.com

-  Oracle: 47% market share, \$7.2BN in sales
-  IBM: 21% market share with \$3.2BN in sales
-  Microsoft: 17% market with \$2.6BN in sales

DBMS Market Shares (Cloud based)

2017		2018		2019		2020		2021	
Vendor	Share	Vendor	Share	Vendor	Share	Vendor	Share	Vendor	Share
Oracle	36.1%	Oracle	31.1%	Oracle	27.4%	Microsoft	24.3%	Microsoft	24.0%
Microsoft	21.5%	Microsoft	23.6%	Microsoft	24.7%	Oracle	23.8%	AWS	23.9%
IBM	12.7%	AWS	13.5%	AWS	17.1%	AWS	20.6%	Oracle	20.6%
AWS	9.2%	IBM	10.4%	IBM	8.8%	IBM	6.8%	Google	6.5%
SAP	7.4%	SAP	6.9%	SAP	6.5%	SAP	5.6%	IBM	5.6%

DBMS Latest ranking

<http://db-engines.com/>

Rank			DBMS
May 2022	Apr 2022	May 2021	
1.	1.	1.	Oracle 
2.	2.	2.	MySQL 
3.	3.	3.	Microsoft SQL Server 
4.	4.	4.	PostgreSQL  
5.	5.	5.	MongoDB 
6.	6.	 7.	Redis 
7.	 8.	 6.	IBM Db2
8.	 7.	8.	Elasticsearch 
9.	9.	 10.	Microsoft Access
10.	10.	 9.	SQLite 

An Example

The Internet Movie Database

<http://www.imdb.com>

- Entities:
Actors (800k), Movies (400k), Directors, ...
- Relationships:
who played where, who directed what, ...

Tables

Directors:

id	fName	lName
15901	Francis Ford	Coppola
...		

Movie_Directors:

id	mid
15901	130128
...	

Movies:

mid	Title	Year
130128	The Godfather	1972
...		

A Motivating, Running Example

- Consider building a course management system (**CMS**):

- Students

- Courses

- Professors



Entities

- Who takes what

- Who teaches what



Relationships

Modeling the CMS using DBMS

- *Logical Schema*

- Students(sid: *string*, name: *string*, gpa: *float*)
- Courses(cid: *string*, cname: *string*, credits: *int*)
- Enrolled(sid: *string*, cid: *string*, grade: *string*)
-

sid	Name	Gpa
101	Jalil	3.2
123	Karim	3.8

Students

Relations

cid	cname	credits
101	C	3
303	Dbase	3

Courses

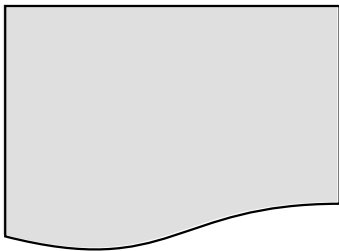
sid	cid	Grade
123	303	A

Enrolled

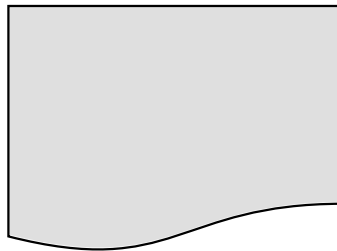
Can we do it without a DBMS ?

Sure we can! Start by storing the data in files:

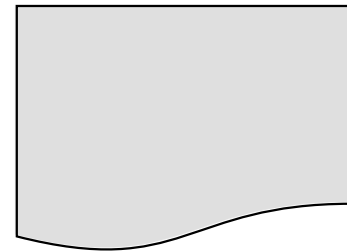
students.txt



courses.txt



professors.txt



You can store data using text editors, OR

You can write C or Java programs to
implement specific tasks

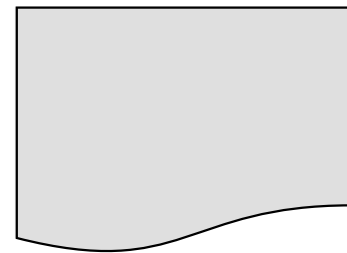
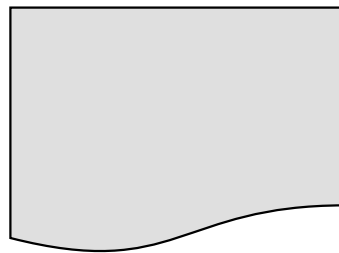
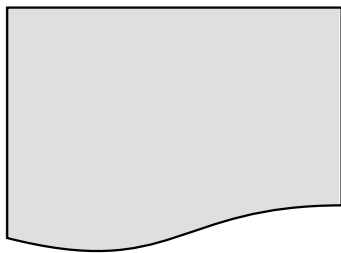
Can we do it without a DBMS ?

Another way: store the data in EXCEL files:

students.xls

courses.xls

professors.xls



Possible Organizations

- Files
- Spreadsheets
- DBMS

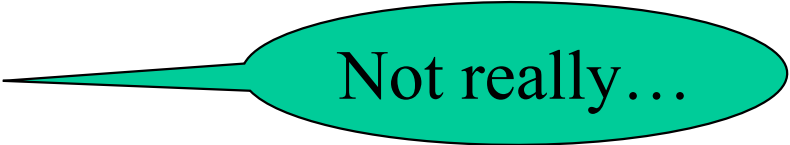
1. Create/store Large Datasets

- Files



Yes, but...

- Spreadsheets



Not really...

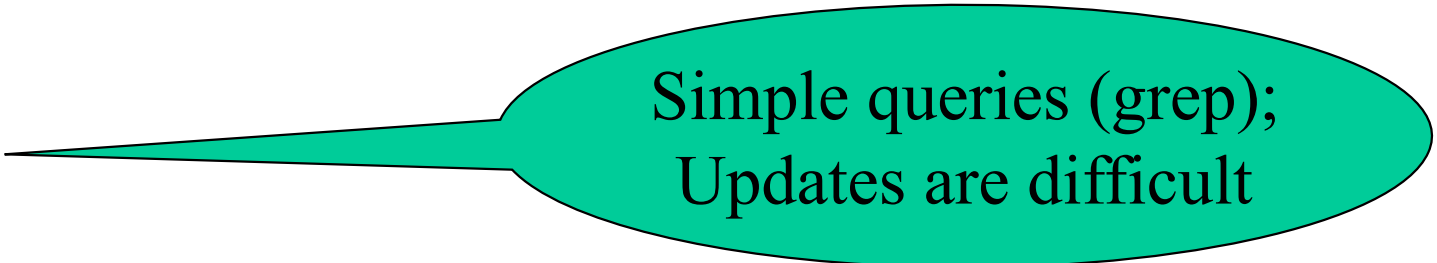
- DBMS



Yes

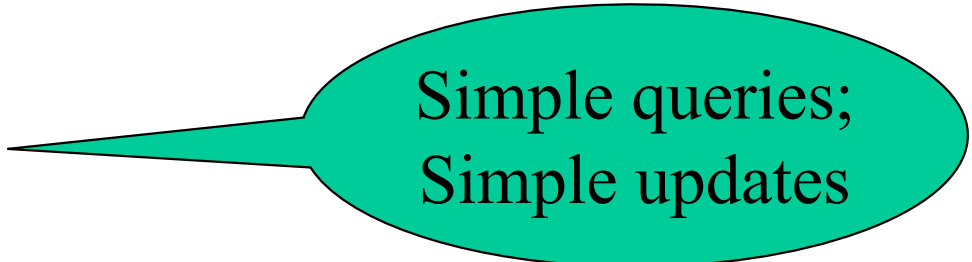
2. Search/Query/Update

- Files



Simple queries (grep);
Updates are difficult

- Spreadsheets



Simple queries;
Simple updates

- DBMS

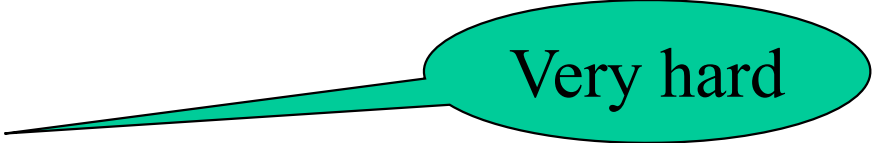


All

3. Change the Structure

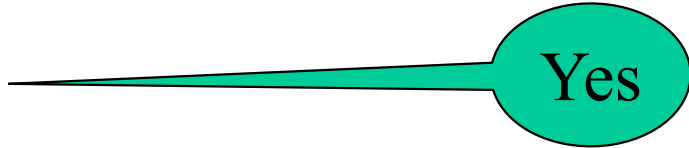
Add Address to each Actor

- Files



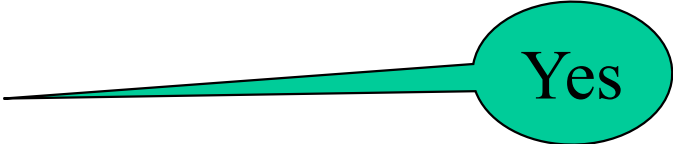
Very hard

- Spreadsheets



Yes

- DBMS



Yes

4. Concurrent Access

Multiple users access/update the data concurrently

- What can go wrong ?

Lost updates; inconsistent reads,...

- You and your project partner are editing the same file.
- You both save it at the same time.
- Whose changes survive?

- How do we protect against that in OS ?

locks

5. Recover from crashes

- Transfer \$100 from account #4662 to #7199:

```
X = Read(Account, #4662);  
X.amount = X.amount - 100;  
Write(Account, #4662, X);
```

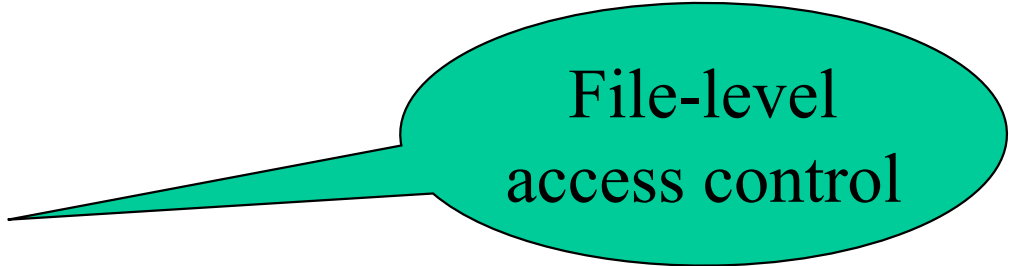
```
Y = Read(Account, #7199);  
Y.amount = Y.amount + 100;  
Write(Account, #7199, Y);
```

CRASH !

What is the problem ?

6. Security

- Files



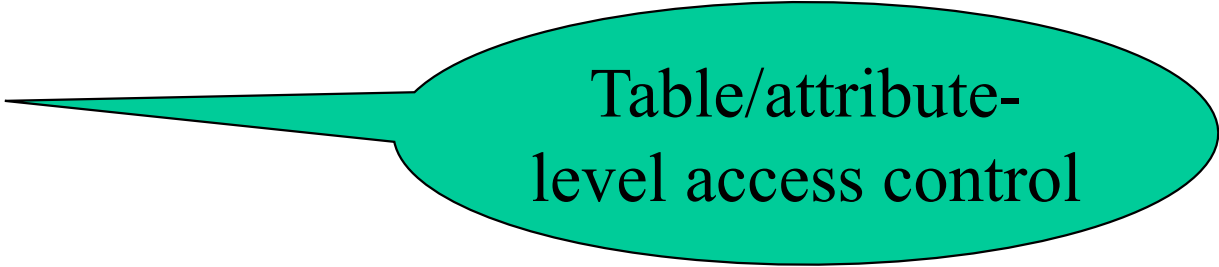
File-level
access control

- Spreadsheets



Same [?]

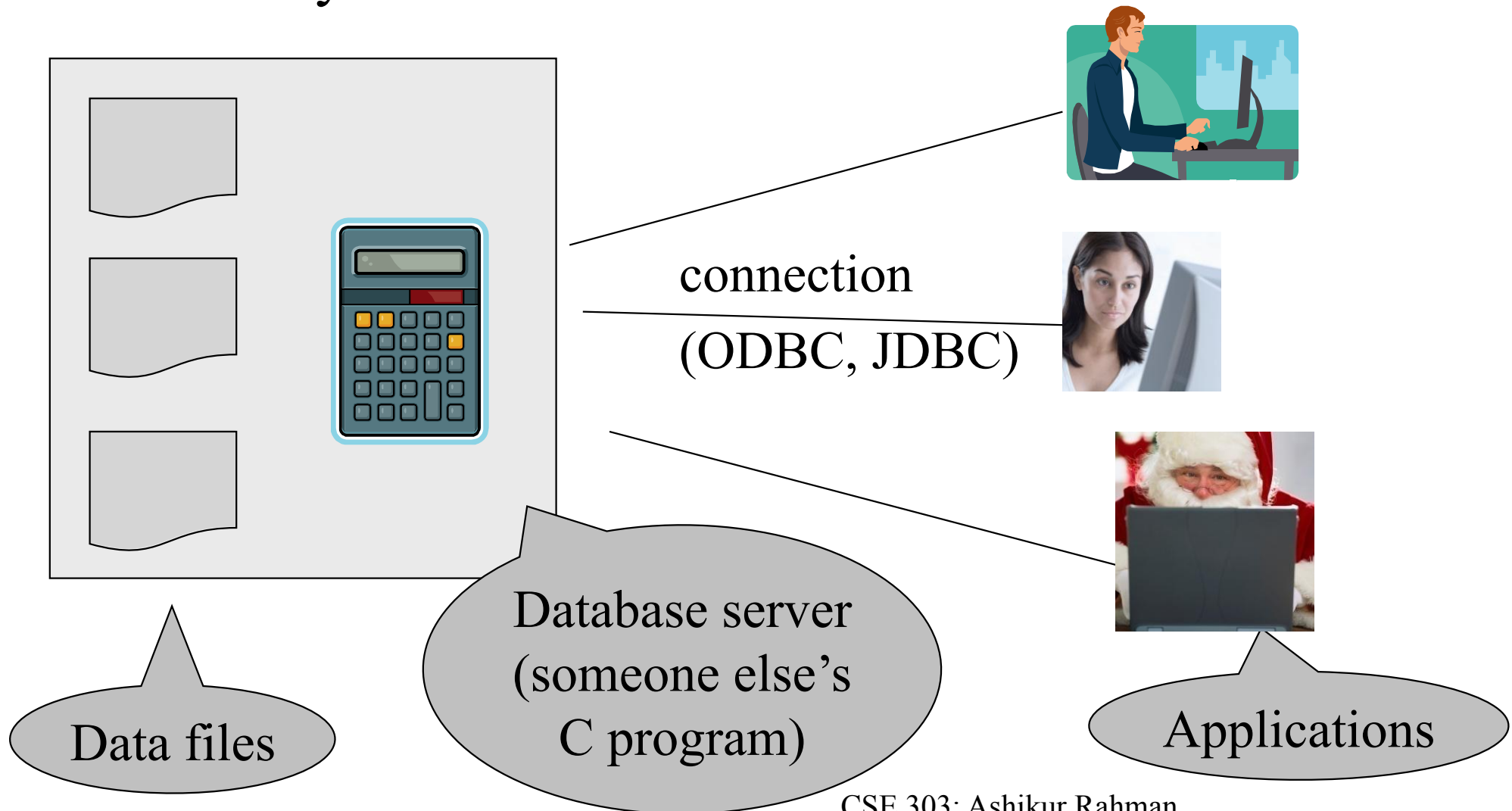
- DBMS



Table/attribute-
level access control

How DBMS does all those things

“Two tier system” or “client-server”



Data Independence

Logical view

Directors:

id	fName	lName
15901	Francis Ford	Coppola
...		

Movie_Directors:

id	mid
15901	130128
...	

Movies:

mid	Title	Year
130128	The Godfather	1972
...		

Directors_file

Moviews_title_index_file

Directors_fname_index_file

Movies_file

Physical view

What the Database Systems Does

-
- The diagram consists of a numbered list of six database functions on the left. To the right of each function is a gray rounded rectangle containing a related concept. A thin gray line with a pointed arrow connects the function to the box. For example, '1. Create/store large datasets' is connected to 'SQL DDL', '2. Search/query/update' to 'SQL DML', '3. Change the structure' to 'SQL DDL', '4. Concurrent access to many user' to 'Transactions ACID', and '6. Security' to 'Grant, Revoke, Roles'. Item 5, 'Recover from crashes', does not have a corresponding box.
1. Create/store large datasets SQL DDL
 2. Search/query/update SQL DML
 3. Change the structure SQL DDL
 4. Concurrent access to many user Transactions
ACID
 5. Recover from crashes
 6. Security Grant, Revoke, Roles

Course Outline (first half) - TENTATIVE !!

1. Introduction (chapter 1)
2. Relational model (chapter 2)
3. SQL (chapter 6)
4. Constraints (chapter 7)
5. Design theory (chapter 3)
6. E-R diagram (chapter 4)
7. Relational Algebra (chapter 2)

Until Next Time...

- Go to the course website
- Install Oracle Xpress Edition 11/12
- Start using sqlplus, and sql developer
- Read the guideline on the course website to use sqlplus
- Find a partner for the project in the lab
- Start reading about SQL online and in the book